

EDUCATION

IMS Engineering College	July 2022 – July 2026
<i>B. Tech (CSE) - CGPA - 8.92</i>	
Delhi Public School Ghaziabad	July 2021 – July 2022
<i>High School - 87%</i>	
St. Joseph's Academy	May 2019 – July 2020
<i>Secondary School - 97%</i>	

EXPERIENCE

Trilogy Innovations  Jan 2026 – Present

SDE Intern

- Leading end-to-end development of an intelligent campus surveillance and face recognition system, owning the complete lifecycle from system design, UI/UX design, and architecture planning through development, testing, and production deployment on AWS.
- Architecting and building a full-stack application using Next.js, FastAPI, and AWS (ECS, Amplify, S3, CodeBuild), integrating real-time camera feeds with deep learning-based face recognition pipelines to enable automated Interaction & presence tracking and visitor management.
- Driving all aspects of delivery as the sole developer, including requirement analysis, database schema design, RESTful API development, responsive frontend implementation, CI/CD pipeline configuration, and iterative QA, translating business requirements into a deployed, production-grade solution.

Amity University, Noida  June 2024 – July 2024

Undergraduate Student Researcher

- Designed and implemented a custom CNN architecture using transfer learning and multi-head attention to classify biodegradable vs. non-biodegradable waste, achieving 95.12% accuracy on a dataset of 76,000+ labeled images.
- Deployed the model to a real-time mobile application using Flutter and TensorFlow Lite, enabling on-device inference for waste classification.
- Published a peer-reviewed [research paper](#) based on this work in IEEE Xplore, detailing the architecture, the dataset, and the deployment strategies.

PROJECTS

LiveTranslate  | Python, Whisper ASR, NLLB, Web Speech API, TTS, FastAPI Oct 2025

- Developed a real-time speech translation system leveraging local LLM pipelines using OpenAI Whisper for streaming ASR, Meta's NLLB for multilingual translation, and Web Speech API for browser-side speech processing.
- Engineered an on-device translation workflow using punctuation-restoration and text-normalisation models, improving transcript readability and translation fluency.

HerShield  | Flutter, Dart, Google Maps API, RAG chatbot Mar 2024

- Engineered a women's safety app with features like safety indexing for localities based on crime data.
- Developed a community-assist mechanism to alert nearby users during reported emergencies using Google Maps API.
- Integrated a Retrieval-Augmented Generation (RAG) chatbot for guidance and emergency procedures.

TECHNICAL SKILLS

- Programming: Data Structures and Algorithms, OOPS, DBMS, OS.
- Languages: Java, C, Python, JavaScript, HTML, CSS, MySQL.
- Frameworks: ReactJS (Library), NodeJS, Express, Zod, Mongoose, MongoDB Atlas, Tailwind CSS, Bootstrap.
- Technologies: Firebase, Pinecone, LangChain, LangGraph, Google.GenAI, Google Maps API, FastAPI, tflite, Flutter (basic).
- Skills: Deep Learning, Transfer Learning, RAG, LLM Integration, Agentic AI(Agent SDK), APIs, Web Dev., Git, GitHub, Claude skills, AWS(S3, ECS, Fargate, TimeStream, DynamoDB, Codebuild, ElasticCache).

ACHIEVEMENTS/CERTIFICATIONS

- 1st Prize – IEEE Mega Project 7.0, Birla Institute of Technology (2025) 
- 1st Prize – BugSense AI Bug Hunting, IMS UC Campus (2025) 
- 2nd Prize – Hackmatrix Hackathon, IIT Patna (2024) 
- Certificate of Appreciation – Walmart Sparkathon (2024) 
- Certified in Deep Learning – NVIDIA (2024) 